Supplemental Online Content

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This supplemental material has been provided by the authors to give readers additional information about their work.

SUPPLEMENT 3

eMethods

Consent model

The trial was conducted using a hybrid model of consent: 1) enrollment using a full waiver of consent at Australian sites; and 2) enrollment with a waiver of consent to commence the intervention using a treating clinician "best interest statement" with relative/Whanau/friend consultation, when possible, with consent to continue participation obtained from the patient or their medical treatment decision maker at the New Zealand site. When patients or their medical treatment decision maker opted out or refused consent to continue participation, permission was requested to retain data already collected.

Enteral formula product information

Nutrison Protein Plus: https://nutricia.com.au/adult/product/nutrison-protein-plus/
Nutrison Protein Intense: https://nutricia.com.au/adult/product/nutrison-protein-intense/

Ideal body weight calculation

While enteral nutrition delivery rates were conducted as per usual site practice, study sites received education on calculating ideal body weight for overweight and obese patients when prescribing nutrition and were recommended to prescribe no more than a maximum of 1ml per kg of ideal body weight per hour. Ideal body weight was calculated from patient height per site current practice.

Calculation of duration of trial enteral formula

Nutrition intake data was collected on days 1-5, day 10, day 20, day 30 and day 90. Date and time of the commencement of trial enteral formula and the date and time of trial enteral formula cessation for the last time in the index ICU admission were recorded. Readmissions to ICU during the index hospital admission up to 90 days post commencement of trial enteral nutrition were recorded. If the trial enteral nutrition was re-commenced during the readmission to ICU, the date and time of the re-commencement of trial enteral nutrition and the date and time of trial enteral nutrition cessation for the last time in the ICU re-admission were recorded. Duration of trial enteral nutrition was calculated in hours from the time of commencement of the trial enteral nutrition until cessation of the last episode of trial enteral nutrition. Duration of trial enteral nutrition was calculated in hours to accurately represent the duration of feeding because there are at least two partial days (day of commencement of trial enteral nutrition and day of ceasing trial enteral nutrition) of feeding and presenting as feeding days would inflate the estimate of trial enteral nutrition delivered.

Trial enteral nutrition delivery audits

A total of eight trial enteral nutrition audits were conducted during the trial. These audits occurred at the following timepoints: 1) **mid cluster,** i.e. 6 weeks after each cluster period commencement (four timepoints); 2) **at cluster change over,** i.e. in the week following cluster change over (three timepoints) 3) **at study completion**, i.e., at one timepoint within one week of study completion. Audit dates were randomly assigned during the audit week to sites, excluding weekends, using an online randomizer - https://www.random.org/calendar-dates/. The time of the day in which the audit took place was selected by sites for convenience, but the entire audit was instructed to be completed at once. For the audit, sites visualized and recorded in the electronic case report form the trial formula hanging at the bedside. During data analysis, the recorded formula hanging at the bedside was compared to the trial enteral nutrition the patient was initially assigned to receive. These were done to identify patients who may be receiving the alternate trial enteral nutrition or a non-trial formula. We also identified and counted patients who were in the ICU but were not receiving enteral nutrition at time of the audit i.e. fasting or ceased clinical need for enteral nutrition.

Duration of invasive ventilation

Duration of invasive ventilation was considered as a lower bounded continuous response, rather than as a time-to-event outcome as stated in the published statistical analysis plan. A tobit mixed effects model[3] was used to compare the mean duration of invasive ventilation between treatment groups and included covariate adjustment for period and delayed start. ICU was included as a random effect and assumed to be normally distributed with mean zero and variance component σ_C^2. The treatment effect estimate from the tobit mixed effects model is presented as a difference in means (95% CI).

Adverse events and serious adverse events

It is recognized that the patient population in the ICU experience aberrations in laboratory values, signs and symptoms due to the severity of the underlying disease and the impact of standard treatments in the ICU. These do not necessarily constitute adverse events or serious adverse events unless they are considered related to study treatment or, in the principal investigator's clinical judgement, are not recognized events consistent with the patient's underlying critical illness and/or chronic diseases and expected clinical course. Accordingly, the reporting of adverse events (AEs) was restricted to events that were considered related to study treatment (possibly, probably or definitely). AEs were defined as any untoward medical occurrence in a patient or clinical investigation subject administered an investigational intervention and were collected from commencement of trial enteral formula until 48 hours post cessation of trial enteral formula. The patient was followed until the event was resolved or explained. Frequency of follow-up was left to the discretion of the site principal investigator.

The baseline mortality of intensive care patients is high due to the illness that has necessitated admission. ICU patients will frequently develop life-threatening organ failure(s) unrelated to study interventions and despite optimal management. Therefore, consistent with ICU international best practice and, as outlined above, deaths that were part of the natural history of the primary disease process or expected complications of critical illness were not reported as serious adverse events (SAEs) in this study[4]. Events already defined and reported as study outcomes (e.g., mortality) were not reported separately as AEs and SAEs unless they were considered to be causally related to the study intervention or were otherwise of concern in the site principal investigator's judgement. SAEs were reported up to day 90 post-study enrolment, including SAEs that were considered related to trial enteral nutrition, namely mesenteric ischemia and abdominal compartment syndrome, were at the discretion of each site's principal investigator[5, 6].

Tertiary outcomes

Biochemical data were collected on study days 1, 5 and 10 only whilst in ICU.

Secondary analyses

Secondary analyses for the primary outcome were conducted by fitting 1) a linear mixed model to align with the sample size calculation, and 2) a Bayesian quantile mixed model to "augment" interpretation of the trial by producing a distribution of effect sizes that are compatible with the observed trial data (posterior distribution) and the probability of any treatment benefit. To summarize the posterior distribution, the posterior mean of the regression coefficient for treatment from the Bayesian quantile mixed effects model (difference in medians) and 95% credible interval (2.5th and 97.5th percentiles of the posterior distribution) are presented.

Sensitivity analyses

Planned sensitivity analyses for the primary outcome were conducted excluding patients known to have received non-trial enteral nutrition and patients admitted for palliative care or organ donation. A sensitivity analysis was planned using imputation for missing primary outcome data but was not required because there were no missing data.

R packages and Stata commands

Primary analysis

Quantile mixed effects modelling was performed using the lqmm function in R's lqmm package [7, 8]. The derivative free (Nelder-Mead) estimation method was specified, and default initial values were used (least squares estimate for the fixed effects and a value of one for the variance component). 95% CIs were generated from 1000 bootstrap samples/replications.

A model-based residual variance can be derived as described in[7] (page 9), and used, along with the estimate of σ_C^2, to estimate the intracluster correlation (ICC).

Secondary analyses

Linear mixed modelling was performed using the lmer function in R's lme4 package[9]. The ICC was derived from the estimate of σ_C^2 and the residual error variance.

Bayesian quantile mixed modelling was performed using the brm function in R's brms package[10, 11]. The default initial values (randomly generated by the brm function) and the default prior distributions were specified:

- intercept: student-t distribution with 3 degrees of freedom, location parameter set to the median of the outcome and scale parameter determined from the standard deviation of the outcome
- fixed effects/regression coefficients: flat (non-informative) priors
- variance component σ_C^2 and residual standard deviation: half student-t distribution with 3 degrees of freedom, location parameter set to 0 and scale parameter determined from the standard deviation of the outcome

Four chains were implemented, and 1000 posterior samples were retained from each chain after a burn-in of 1000 iterations, resulting in 4000 samples per parameter for the calculation of posterior summaries (mean and 95% credible interval, which is calculated from the 2.5th and 97.5th percentiles of the posterior samples). Trace plots and standard Markov chain Monte Carlo (MCMC) diagnostic criteria, including the potential scale reduction ($\hat{R}$) statistic and the effective sample sizes (ESS), were used to check convergence.

For each iteration post burn-in, a model-based residual variance was derived as described in[7] (page 9), and then σ_C^2 was divided by the sum of the model-based residual variance and σ_C^2. The mean of the resulting values was used to estimate the intracluster correlation (ICC).

Alive at day 90, incidence of tracheostomy insertion and incidence of kidney replacement therapy that commenced after trial enteral nutrition

Fay and Graubard's small sample bias correction of the sandwich variance estimator can be implemented using Stata's xtgeebcv command [12, 13]. ICC estimates were obtained using the postestimation command estat wcorrelation after xtgee fit (i.e., from the model without the small sample bias correction applied), as the xtgeebcv command does not appear to be compatible with the postestimation estat wcorrelation command.

Duration of admission to ICU and admission to hospital

The cumulative incidence function was estimated using the cuminc function in R's cmprsk package[14, 15]. Cause-specific Cox regression analyses were performed using the stcox command in Stata.

Duration of invasive ventilation

Tobit mixed effects modelling was performed using the metobit function in Stata, which by default, reports effect estimates for the latent variable assumed by the tobit model. Stata's margins function, with dydx(treatment variable) and predict(ystar(0,.)) was used to derive an estimate of the treatment effect on the observed outcome. An estimate of the ICC was obtained using the postestimation command estat icc.

Subgroup analyses

P-values for the test of interaction between treatment and each subgroup was derived from a likelihood ratio test comparing the quantile mixed models excluding and including the interaction term using the lrtest function from R's lmtest package[13].

A subgroup specific treatment effect estimate for the primary outcome was obtained from the relevant coefficient estimates from the interaction model (treatment (main) effect estimate for reference subgroup and treatment (main) effect estimate plus interaction term estimate for the non-reference subgroup) and corresponding 95% CIs were calculated using the block-bootstrap method.

eTables

eTable 1: Patient inclusion and exclusion criteria

Inclusion criteria	
1.	≥16 years of age
2.	Receiving/about to commence enteral nutrition during the index admission to ICU or receiving/about to commence enteral nutrition for the first time in ICU during subsequent admissions
Exclusion criteria	
1.	At the commencement of enteral nutrition, the treating clinician considers the trial enteral formula to be contraindicated
2.	The patient has received greater than 12 hours of non-trial enteral formula
3.	The patient has previously participated in the TARGET Protein trial

ICU = Intensive Care Unit

eTable 2: Outcome definitions

Primary outcome
• Number of days free of the index hospital and alive at day 90. Calculated as 90 days minus all days admitted to the index hospital after commencement of trial enteral formula minus any days readmitted to the index hospital within 90 days. As patients may be readmitted for part of a day (e.g. for dialysis) a readmission was only be counted when the patient remained in hospital at midnight. Patients were considered alive if they survive the index-hospital episode, and there is no evidence of death before day 90. In Australia, death after hospital discharge was ascertained from the local health record by site investigators and using data linkage with the Australian National Death Index. At the New Zealand site, the Australian New Zealand Intensive Care Society Adult Patient Database (ANZICS APD) is linked to the New Zealand national death registry to record death after hospital discharge. Patients who die during the 90-day post-trial recruitment period, were assigned a days free of the index hospital value of zero.
Secondary outcomes
• Days free of the index hospital and alive at day 90 in survivors. Calculated only for those that survived to day 90. Calculated the same as the primary outcome, i.e., 90 days minus all days admitted to the index hospital after commencement of trial enteral formula minus any days readmitted to the index hospital within 90 days. As patients may be readmitted for part of a day (e.g., for dialysis) a readmission was only be counted when the patient remained in hospital at midnight. Patients who remained in hospital until day 90 were assigned a value of zero. • Alive at day 90. Patients were considered alive if they were alive at discharge from the index hospital, and there was no evidence of death before day 90. Local health service records were used to provide evidence of patient status, including if the patient was discharged alive from hospital but known to have a post discharge health event. Health events include rehabilitation facility admission, outpatient appointment attended, pathology test completed, other medical investigation, or day admission e.g. dialysis, chemotherapy. Binary outcome (yes/no). • Duration of invasive ventilation (hours). Total hours of invasive ventilation for patients receiving invasive ventilation in ICU. A continuous outcome restricted to be non-negative. • Duration of admission to ICU (days). Time-to-ICU discharge with death in ICU considered as a competing risk. • Duration of admission to hospital (days). Time-to-hospital discharge with death in hospital considered as a competing risk. • Incidence of tracheostomy insertion • Incidence of new kidney replacement therapy commenced during index ICU admission, subsequent to commencement of trial enteral nutrition • Hospital discharge destination – Home, other hospital ICU, other acute hospital, rehabilitation, long term care, and other

ICU = Intensive Care Unit

eTable 3: Baseline characteristics of participants by sequence

		Sequence: Augmented Protein / Usual Protein / Augmented Protein / Usual Protein (4 units, n = 2044)	Sequence: Usual Protein / Augmented Protein / Usual Protein / Augmented Protein (4 units, n = 1353)
Region – number of units, n (%)			
	Australia	3 units, n = 1,784 (87.3)	4 units, n = 1,353 (100.0)
	New Zealand	1 unit, n = 260 (12.7)	0 units, n = 0 (0.0)
ICU site – [n (%)]			
	Royal Adelaide Hospital	0 (0.0)	660 (48.8)
	The Queen Elizabeth Hospital	0 (0.0)	63 (4.7)
	Royal Prince Alfred Hospital	625 (30.6)	0 (0.0)
	Wellington Regional Hospital	260 (12.7)	0 (0.0)
	Austin Hospital	0 (0.0)	394 (29.1)
	Sunshine Hospital	246 (12.0)	0 (0.0)
	Royal Melbourne Hospital	913 (44.7)	0 (0.0)
	University Hospital Geelong	0 (0.0)	236 (17.4)

ICU = Intensive Care Unit
All ICUs are tertiary (medical and surgical)
Trauma hospitals: Royal Melbourne Hospital, Royal Adelaide Hospital, Royal Prince Alfred Hospital and Wellington Regional Hospital
Sites commencing 23 May 2022: Royal Adelaide Hospital, The Queen Elizabeth Hospital, Royal Prince Alfred Hospital, Wellington Regional Hospital
Sites commencing 23 August 2022: Austin Hospital, Sunshine Hospital, Royal Melbourne Hospital, University Hospital Geelong

eTable 4: Baseline characteristics of participants by period and treatment group

	Period 1 - Augmented Protein **(4 units, n = 480)**	**Period 1 –** Usual Protein **(4 units, n = 335)**	**Period 2 -** Augmented Protein **(4 units, n = 298)**	**Period 2 –** Usual Protein **(4 units, n = 530)**	**Period 3 -** Augmented Protein **(4 units, n = 551)**	**Period 3 –** Usual Protein **(4 units, n = 368)**	**Period 4 -** Augmented Protein **(4 units, n = 352)**	**Period 4 –** Usual Protein **(4 units, n = 483)**
Sex [n (%)]								
Male	303 (63.1)	207 (61.8)	187 (62.8)	319 (60.2)	359 (65.2)	226 (61.4)	220 (62.5)	335 (69.4)
Female	177 (36.9)	128 (38.2)	111 (37.2)	211 (39.8)	190 (34.5)	142 (38.6)	132 (37.5)	148 (30.6)
Age, years	61.0 (48.0, 71.0)	62.0 (53.0, 73.0)	64.0 (50.0, 72.0)	60.0 (45.0, 71.0)	59.0 (45.0, 69.0)	62.0 (48.0, 72.5)	62.0 (49.0, 71.0)	62.0 (47.0, 71.0)
Weight, kilograms	80.1 (68.0, 95.5)	80.0 (65.0, 95.0)	78.0 (67.9, 94.5)	81.5 (69.0, 99.0)	80.5 (68.0, 94.0)	80.0 (68.0, 98.5)	80.0 (68.0, 93.0)	80.0 (67.0, 92.0)
Body Mass Index, kg/m^2	27.6 (24.1, 33.1)	27.0 (23.7, 32.2)	27.3 (23.8, 31.6)	27.9 (24.2, 33.5)	27.3 (24.4, 32.8)	27.5 (23.9, 32.1)	28.4 (23.9, 32.5)	27.4 (23.4, 31.2)
Ideal body weight, kilograms[1]	65.8 (56.8, 72.1)	66.7 (56.8, 73.0)	65.8 (56.8, 74.8)	65.8 (55.5, 72.1)	65.8 (56.8, 73.0)	65.8 (55.0, 74.8)	65.8 (56.8, 73.0)	65.8 (56.8, 73.0)
APACHE II score at ICU admission	20.0 (16.0, 25.0)	18.0 (14.0, 24.0)	19.0 (14.0, 24.0)	20.0 (15.0, 25.0)	20.0 (15.0, 25.0)	19.0 (14.0, 24.0)	18.0 (14.0, 24.0)	19.0 (14.0, 25.0)
ICU admission [n (%)]								
Emergency surgical	97 (20.2)	73 (21.8)	82 (27.5)	106 (20.0)	123 (22.3)	86 (23.4)	77 (21.9)	96 (19.9)
Elective surgical	66 (13.8)	59 (17.6)	40 (13.4)	75 (14.2)	71 (12.9)	47 (12.8)	57 (16.2)	74 (15.3)
Non-surgical (Medical)	317 (66.0)	203 (60.6)	176 (59.1)	349 (65.8)	357 (64.8)	235 (63.9)	218 (61.9)	313 (64.8)
ICU admission diagnosis category[2] **[n (%)]**								
Cardiovascular	95 (19.8)	62 (18.5)	47 (15.8)	102 (19.2)	106 (19.2)	58 (15.8)	52 (14.8)	88 (18.2)
Respiratory	94 (19.6)	97 (29.0)	77 (25.8)	80 (15.1)	72 (13.1)	96 (26.1)	82 (23.3)	87 (18.0)
Gastrointestinal	44 (9.2)	36 (10.7)	21 (7.0)	56 (10.6)	63 (11.4)	42 (11.4)	40 (11.4)	56 (11.6)
Neurological	113 (23.5)	56 (16.7)	71 (23.8)	118 (22.3)	126 (22.9)	73 (19.8)	64 (18.2)	103 (21.3)
Sepsis	32 (6.7)	24 (7.2)	32 (10.7)	37 (7.0)	46 (8.3)	33 (9.0)	38 (10.8)	40 (8.3)
Trauma	61 (12.7)	29 (8.7)	21 (7.0)	81 (15.3)	84 (15.2)	32 (8.7)	38 (10.8)	67 (13.9)
Other	41 (8.5)	31 (9.3)	29 (9.7)	56 (10.6)	54 (9.8)	34 (9.2)	38 (10.8)	42 (8.7)
Clinical frailty score	3.0 (3.0, 4.0)	3.0 (3.0, 4.0)	3.0 (3.0, 5.0)	3.0 (3.0, 4.0)	3.0 (2.0, 4.0)	3.0 (3.0, 5.0)	3.0 (3.0, 4.0)	3.0 (3.0, 4.0)
Diabetes [n (%)]	122 (25.4)	103 (30.7)	79 (26.5)	115 (21.7)	137 (24.9)	101 (27.4)	91 (25.9)	132 (27.3)

	Period 1 - Augmented Protein **(4 units, n = 480)**	**Period 1 –** Usual Protein **(4 units, n = 335)**	**Period 2 -** Augmented Protein **(4 units, n = 298)**	**Period 2 –** Usual Protein **(4 units, n = 530)**	**Period 3 -** Augmented Protein **(4 units, n = 551)**	**Period 3 –** Usual Protein **(4 units, n = 368)**	**Period 4 -** Augmented Protein **(4 units, n = 352)**	**Period 4 –** Usual Protein **(4 units, n = 483)**
ICU source of admission [n (%)]								
Emergency room	181 (37.7)	103 (30.7)	96 (32.2)	210 (39.6)	197 (35.8)	117 (31.8)	115 (32.7)	188 (38.9)
Hospital ward	71 (14.8)	50 (14.9)	34 (11.4)	63 (11.9)	63 (11.4)	67 (18.2)	59 (16.8)	63 (13.0)
Operating theatre	144 (30.0)	130 (38.8)	120 (40.3)	170 (32.1)	180 (32.7)	131 (35.6)	131 (37.2)	164 (34.0)
Transfer from another ICU	32 (6.7)	16 (4.8)	15 (5.0)	35 (6.6)	48 (8.7)	15 (4.1)	14 (4.0)	26 (5.4)
Other	52 (10.8)	36 (10.7)	33 (11.1)	52 (9.8)	63 (11.4)	38 (10.3)	33 (9.4)	42 (8.7)
Invasively ventilated at trial enteral nutrition commencement [n (%)]	398 (82.9)	261 (77.9)	233 (78.2)	429 (80.9)	476 (86.4)	285 (77.4)	278 (79.0)	380 (78.7)
Received vasopressor or inotrope at trial enteral nutrition commencement [n (%)]	274 (57.1)	172 (51.3)	153 (51.3)	286 (54.0)	307 (55.7)	182 (49.5)	192 (54.5)	281 (58.2)
New kidney replacement therapy at trial enteral nutrition commencement [n (%)]	42 (8.8)	28 (8.4)	23 (7.7)	31 (5.8)	29 (5.3)	31 (8.4)	28 (8.0)	29 (6.0)
Received parenteral nutrition [n (%)]	41 (8.5)	12 (3.6)	11 (3.7)	44 (8.3)	37 (6.7)	23 (6.2)	26 (7.4)	41 (8.5)
Blood urea, mmol/l	7.9 (5.1, 13.1)	7.7 (5.3, 12.7)	7.4 (5.3, 11.9)	7.3 (5.0, 11.4)	7.2 (5.4, 12.0)	7.9 (5.2, 12.6)	7.6 (5.1, 12.4)	7.9 (5.2, 12.9)
Highest serum creatinine, µmol/l	93.0 (67.0, 159.0)	97.0 (66.0, 153.0)	95.5 (68.0, 146.0)	90.0 (67.0, 138.0)	89.0 (66.0, 144.0)	96.0 (67.5,167.5)	98.0 (74.0,150.0)	95.5 (69.0,157.0)
Lowest serum creatinine, µmol/l	81.0 (61.0, 128.0)	85.0 (61.0, 130.0)	83.5 (62.0, 130.0)	78.0 (58.0, 115.0)	78.0 (59.0, 115.0)	84.5 (62.0, 131.0)	82.0 (66.0, 125.0)	84.0 (64.0, 124.0)
Highest serum albumin, g/l	31.0 (27.0, 34.0)	30.0 (26.0, 34.0)	30.0 (25.0, 34.0)	31.0 (27.0, 35.0)	31.0 (27.0, 35.0)	30.0 (25.0, 34.0)	30.0 (26.0, 34.0)	31.0 (27.0, 34.0)

	Period 1 - Augmented Protein (4 units, n = 480)	Period 1 – Usual Protein (4 units, n = 335)	Period 2 - Augmented Protein (4 units, n = 298)	Period 2 – Usual Protein (4 units, n = 530)	Period 3 - Augmented Protein (4 units, n = 551)	Period 3 – Usual Protein (4 units, n = 368)	Period 4 - Augmented Protein (4 units, n = 352)	Period 4 – Usual Protein (4 units, n = 483)
Lowest serum albumin, g/l	28.0 (24.0, 31.0)	28.0 (23.0, 31.0)	27.0 (23.0, 31.0)	28.0 (24.0, 32.0)	28.0 (24.0, 32.0)	27.0 (23.0, 31.0)	28.0 (23.0, 31.0)	28.0 (24.0, 32.0)
Highest serum potassium, mmol/l	4.4 (4.0, 4.9)	4.5 (4.1, 5.0)	4.4 (4.0, 4.8)	4.4 (4.0, 4.8)	4.4 (4.0, 4.8)	4.4 (4.0, 4.9)	4.4 (4.0, 4.8)	4.5 (4.1, 5.0)
Lowest serum potassium, mmol/l	3.9 (3.6, 4.3)	4.1 (3.8, 4.6)	4.0 (3.7, 4.4)	3.9 (3.5, 4.2)	3.8 (3.5, 4.2)	4.1 (3.7, 4.4)	4.0 (3.7, 4.3)	3.9 (3.5, 4.3)
Highest blood glucose, mmol/l	10.4 (8.5, 14.3)	9.8 (7.7, 12.7)	8.8 (7.2, 11.6)	10.3 (8.5, 13.4)	10.0 (8.2, 13.1)	9.1 (7.4, 12.1)	9.2 (7.2, 12.0)	10.2 (8.2, 13.8)
Lowest blood glucose, mmol/l	6.5 (5.5, 7.6)	6.9 (5.8, 8.3)	6.9 (5.8, 8.5)	6.4 (5.4, 7.4)	6.5 (5.3, 7.4)	6.9 (5.5, 8.4)	6.6 (5.5, 7.9)	6.4 (5.3, 7.5)
Treatment goals on ICU admission [n (%)]								
Full active management	430 (89.6)	315 (94.0)	283 (95.0)	481 (90.8)	505 (91.7)	348 (94.6)	336 (95.5)	443 (91.7)
Treatment limitation order[3]	42 (8.8)	20 (6.0)	14 (4.7)	42 (7.9)	39 (7.1)	20 (5.4)	15 (4.3)	37 (7.7)
Palliative care of a dying patient	5 (1.0)	0 (0.0)	0 (0.0)	1 (0.2)	0 (0.0)	0 (0.0)	1 (0.3)	1 (0.2)
Potential organ donation	3 (0.6)	0 (0.0)	1 (0.3%)	6 (1.1)	7 (1.3)	0 (0.0)	0 (0.0)	2 (0.4)

APACHE II = Acute Physiology and Chronic Health Evaluation II; ICU = Intensive Care Unit

[1]Calculated from patient height.

[2]Categorized according to APACHE-IIIj: diagnosis which best describes the reason for the ICU admission.

[3]Treatment order implies medical treatment would be constrained at time of ICU admission by patient wishes or likely to be more burdensome than effective (e.g., not for cardiopulmonary resuscitation, not for intubation, not for kidney replacement therapy, maximum dose of vasopressor), but does not necessarily imply an expectation of death during this ICU admission.

eTable 5: Trial enteral formula delivery

Measure	Augmented Protein (n = 1681)	Usual Protein (n = 1716)
Time from ICU admission to trial enteral formula commencement, hours	19.0 (9.2, 37.7)	19.3 (9.5, 39.8)
Duration of trial nutrition, hours	87.0 (36.4, 187.0)	84.0 (34.0, 182.4)
Mean volume of trial enteral formula delivery, ml/day	696.0 (408.0, 950.8)	676.2 (405.0, 956.7)
Calories delivered – kcal/day		
Day 1	353 (176, 605); n =1680	325 (164, 600); n =1711
Day 2	991 (559, 1487); n = 1584	950 (550, 1463); n = 1608
Day 3	1150 (655, 1588); n = 1229	1180 (638, 1650); n = 1247
Day 4	1210 (725, 1663); n = 1005	1219 (700, 1711); n = 998
Day 5	1285 (813, 1707); n = 817	1350 (760, 1733); n = 823
Day 10	1361 (832, 1814); n = 341	1500 (1013, 1925); n = 347
Day 20	1681 (1361, 1966); n = 100	1725 (1075, 2100); n = 91
Day 30	1814 (1191, 2192); n = 43	1825 (1350, 2166); n = 34
Day 90	n = 0	350 (350, 350); n = 1
Protein delivered – g/day		
Day 1	28.0 (14.0, 48.0); n = 1680	16.4 (8.3, 30.2); n = 1711
Day 2	78.7 (44.4, 118.0); n = 1584	47.9 (27.7, 73.7); n = 1608
Day 3	91.3 (52.0, 126.0); n = 1229	59.5 (32.1, 83.2); n = 1247
Day 4	96.0 (57.5, 132.0); n = 1005	61.4 (35.3, 86.2); n = 998
Day 5	102.0 (64.5, 135.5); n = 817	68.0 (38.3, 87.3); n = 823
Day 10	108.0 (66.0, 144.0); n = 341	75.6 (51.0, 97.0); n = 347
Day 20	133.4 (108.0, 156.0); n = 100	86.9 (54.2, 105.8); n = 91
Day 30	144.0 (94.5, 174.0); n = 43	92.0 (68.0, 109.1); n = 34
Day 90	n = 0	17.6 (17.6, 17.6); n = 1
Parenteral nutrition administered at any stage during ICU admission, n (%)	115/1681 (6.8)	120/1716 (7.0)
Protein supplement administered at any stage during ICU admission, n (%)	12/1681 (0.7)	10/1716 (0.6)

Data are median (IQR) or n (%); g = grams; ICU = intensive care unit; Kcal = kilocalories; ml = milliliters

eTable 6: Trial enteral formula delivery - all feeding days

Characteristic	Augmented Protein N = 1,681[1]	Usual Protein N = 1,716[1]
Calories (kcal/kg (IBW)/day)	14 (9, 19)	14 (8, 19)
Missing	369	410
Protein (g/kg (IBW)/day)	1.12 (0.71, 1.53)	0.69 (0.40, 0.95)
Missing	369	410
Calories (kcal/kg (ABW)/day)	10.9 (6.4, 14.9)	10.5 (6.3, 14.9)
Missing	215	248
Protein (g/kg (ABW)/day)	0.87 (0.51, 1.18)	0.53 (0.32, 0.75)
Missing	215	248

[1]Median (Q1, Q3)

IBW - ideal body weight; ABW - actual body weight

103 (49 Treatment – augmented protein and 54 control – usual protein) patients only had volume of enteral nutrition at enrolment and were excluded.

eTable 7: Trial enteral formula delivery - excluding day of enrolment

Characteristic	Augmented Protein N = 1,632[1]	Usual Protein N = 1,662[1]
Calories (kcal/kg (IBW)/day)	17 (11, 22)	17 (10, 22)
Missing	382	425
Protein (g/kg (IBW)/day)	1.33 (0.88, 1.78)	0.84 (0.49, 1.11)
Missing	382	425
Calories (kcal/kg (ABW)/day)	13 (8, 17)	13 (8, 17)
Missing	243	274
Protein (g/kg (ABW)/day)	1.04 (0.63, 1.38)	0.64 (0.40, 0.88)
Missing	243	274

[1]Median (Q1, Q3)

IBW - ideal body weight; ABW - actual body weight

103 (49 Treatment – augmented protein and 54 control – usual protein) patients only had volume of enteral nutrition at enrolment and were excluded.

eTable 8: Protocol deviations

	Augmented Protein (n = 1681)	Usual Protein (n = 1716)
Total protocol deviations, n (%)		
Number of participants with at least one event	151 (9.4)	95 (5.6)
Number of events	158	99
Clinical decision[1], n (%)		
Number of participants with at least one event	92 (5.5)	64 (3.7)
Number of events	96	68
Error on subsequent enteral nutrition administration, n (%)		
Number of participants with at least one event	34 (2.0)	17 (1.0)
Number of events	37	17
Other, n (%)		
Number of participants with at least one event	13 (0.8)	3 (0.2)
Number of events	13	3
Error at initial commencement of enteral nutrition[3], n (%)		
Number of participants with at least one event	11 (0.7)	10 (0.6)
Number of events	11	10
Unable to locate trial enteral formula[4], n (%)		
Number of participants with at least one event	1 (<0.1)	1 (<0.1)
Number of events	1	1

[1] Clinical need for a non-trial enteral nutrition as determined by the treating team

[2] Patients received non-trial enteral nutrition in error after commencing the allocated trial enteral nutrition

[3] Patients received non-trial enteral nutrition in error for less than 12 hours prior to commencing the allocated trial enteral nutrition

[4] Clinical team were unable to locate trial enteral nutrition and delivered non-trial enteral nutrition

eTable 9: Audit of trial enteral formula delivery

	Audit Period			
	Mid-cluster[1]	**Cluster crossover[2]**	**Trial completion[3]**	**Total**
Audit enteral nutrition Type, n (%)				
Correct trial enteral nutrition	188 (64)	153 (68)	35 (67)	376 (66)
Alternate trial enteral nutrition [4]	0 (0)	2 (0.9)	2 (3.8)	4 (0.7)
Other enteral nutrition [5]	15 (5.1)	5 (2.2)	1 (1.9)	21 (3.7)
Not receiving enteral nutrition (fasted, no longer required, etc.)	88 (30)	64 (29)	14 (27)	166 (29)
Missing audit enteral nutrition	1 (0.3)	0 (0)	0 (0)	1 (0.2)
Total	**292 (100)**	**224 (100)**	**52 (100)**	**568 (100)**

[1]Pooled from 4 mid-cluster audits for each hospital

[2]Pooled from 3 cluster crossover audits for each hospital

[3]Audit performed at trial completion

[4]All were protocol deviations (two of which were clinical decisions)

[5]All were protocol deviations (recorded as clinical decision), except for one

568 episodes of auditing of formula delivery of enrolled patients in ICU at the time of audit were conducted. Of these episodes there was only one case of another formula being administered which had not been documented as a protocol deviation and therefore we were unaware of.

Audits of hospitals with delayed trial commencement performed at different times to those that commenced earlier

Other enteral nutrition = any other non-trial formula

Not receiving enteral nutrition = patient in ICU, but not receiving enteral nutrition at time of the audit i.e. fasting or ceased clinical need for enteral nutrition

eTable 10: Summary statistics of the primary and secondary outcome according to period and treatment group

	Period 1 – **Augmented Protein** (4 units, n = 480)	Period 1 – **Usual Protein** (4 units, n = 335)	Period 2 – **Augmented Protein** (4 units, n = 298)	Period 2 – **Usual Protein** (4 units, n = 530)	Period 3 – **Augmented Protein** (4 units, n = 551)	Period 3 – **Usual Protein** (4 units, n = 368)	Period 4 – **Augmented Protein** (4 units, n = 352)	Period 4 – **Usual Protein** (4 units, n = 483)
Number of days free of the index hospital and alive at day 90	61.0 (0.0, 77.0)	64.0 (8.0, 76.0)	63.0 (0.0, 76.0)	65.0 (0.0, 78.0)	62.0 (0.0, 79.0)	61.0 (0.0, 76.0)	63.0 (0.0, 75.5)	65.0 (0.0, 78.0)
Days free of the index hospital at day 90 in survivors	74.0 (61.0, 81.0)	70.0 (57.0, 79.0)	71.0 (57.0, 79.0)	73.0 (60.0, 81.0)	72.0 (55.5, 81.0)	70.5 (58.0, 78.0)	71.0 (56.0, 78.0)	72.0 (60.0, 81.0)
Alive at day 90 [n (%)]	323 (67.3%)	258 (77.0%)	229 (76.8%)	383 (72, 3%)	408 (74.0%)	270 (73.4%)	261 (74.1%)	358 (74.1%)
Duration of admission to ICU (days)	5.1 (2.6, 9.0)	5.8 (3.0, 9.9)	5.1 (3.0, 9.8)	4.4 (2.4, 8.9)	5.1 (2.4, 10.0)	5.3 (2.8, 10.1)	5.1 (2.9, 10.0)	4.4 (2.5, 8.9)
Duration of admission to hospital discharge (days)	11.8 (5.9, 21.1)	15.6 (8.6, 27.8)	14.0 (8.0, 28.5)	12.0 (5.7, 22.2)	12.3 (5.9, 25.9)	14.9 (7.9, 27.2)	15.1 (8.0, 26.7)	11.8 (5.8, 22.8)
Hours of invasive mechanical ventilation in ICU	77.0 (33.2, 166.9)	88.5 (36.0, 184.0)	86.0 (34.0, 180.0)	72.0 (32.0, 148.0)	83.5 (35.7, 185.0)	90.0 (38.0, 177.0)	90.9 (38.3, 185.1)	68.0 (31.0, 138.2)
Tracheostomy in ICU [n (%)]	27 (5.6)	36 (10.7)	29 (9.7)	24 (4.5)	35 (6.4%)	38 (10.3%)	43 (12.2%)	23 (4.8%)
New kidney replacement therapy commenced during index ICU admission after commencing trial enteral nutrition	33 (6.9)	26 (7.8)	22 (7.4)	35 (6.6)	38 (6.9%)	33 (9.0%)	29 (8.2%)	33 (6.8%)
ICU discharge destination [n (%)]								
Died	102 (21.2)	46 (13.7)	43 (14.4)	89 (16.8)	91 (16.5)	64 (17.4)	51 (14.5)	88 (18.2)
Survived ICU	342 (71.2)	277 (82.7)	242 (81.2)	405 (76.4)	409 (74.2)	292 (79.3)	288 (81.8)	357 (73.9)
Transferred to other ICU	4 (0.8)	11 (3.3)	12 (4.0)	13 (2.5)	21 (3.8)	9 (2.4)	11 (3.1)	12 (2.5)
Transferred to other hospital	32 (6.7)	1 (0.3)	1 (0.3)	23 (4.3)	30 (5.4)	3 (0.8)	2 (0.6)	26 (5.4)
At least one hospital readmission (pre-day 90), n (%)	47/480 (9.8)	29/335 (8.7)	12/298 (4.0)	50/530 (9.4)	76/551 (13.8)	26/368 (7.1)	26/352 (7.4)	67/483 (13.9)
Readmission to ICU, days	7.0 (3.0, 17.5)	5.0 (3.0, 10.0)	6.0 (2.0, 12.5)	6.0 (3.0, 11.0)	6.0 (3.0, 11.0)	6.5 (4.0, 10.0)	7.0 (4.2, 10.8)	8.0 (4.0, 13.5)

Abbreviations: ICU = Intensive Care Unit

Data presented as median (IQR) or n (%).

eTable 11: Biochemistry

Measure	Augmented Protein (n = 1681)	Usual Protein (n = 1716)
Blood urea, mmol/l		
Day 1	7.3 (4.8, 12.4); n = 1583	7.5 (4.8, 11.8); n = 1663
Day 5	10.3 (6.8, 16.2); n = 1003	8.4 (5.5, 13.8); n = 985
Day 10	13.0 (8.2, 18.8); n = 439	10.6 (7.1, 15.4); n = 417
Serum albumin, g/l		
Day 1	29.0 (24.0, 33.0); n = 1577	28.0 (24.0, 33.0); n = 1622
Day 5	24.0 (21.0, 27.2); n = 1000	24.0 (21.0, 27.0); n = 979
Day 10	24.0 (21.0, 28.8); n = 438	23.0 (20.0, 28.0); n = 417
Serum phosphate, mmol/l		
Day 1	1.2 (0.9, 1.5); n = 1552	1.2 (0.9, 1.5); n = 1642
Day 5	1.0 (0.9, 1.2); n = 995	1.1 (0.9, 1.3); n = 971
Day 10	1.1 (1.0, 1.4); n = 433	1.2 (1.0, 1.4); n = 408
Blood glucose, mmol/l		
Day 1	7.7 (6.4, 9.6); n = 1662	7.8 (6.4, 9.7); n = 1704
Day 5	7.5 (6.4, 9.1); n = 993	7.8 (6.5, 9.9); n = 992
Day 10	7.5 (6.4, 9.0); n = 442	8.1 (6.8, 10.0); n = 430

Data are Median (IQR); Abbreviations: g = grams; l = liters; mmol/l = millimoles per liter

eTable 12: Serious adverse events and adverse events by treatment group

	Augmented Protein (8 units, n = 1681)	Usual Protein (8 units, n = 1716)
Adverse Event, n (%)		
Number of participants with at least one event	3 (0.2)	1 (0.1)
Number of events	3	1
	(n=1 elevated urea, n=2 *Bacillus Cereus* infection)	(n=1 *Campylobacter* infection)
Serious Adverse Event, n (%)		
Number of participants with at least one event	1 (0.1)	1 (0.1)
Number of events	1	1
	(n=1 *Bacillus Cereus* infection)	(n=1 mesenteric ischemia)

eTable 13: Serious adverse events and adverse events by period

	Period 1 – **Augmented Protein** (4 units, n = 480)	Period 1 – **Usual Protein** (4 units, n = 335)	Period 2 – **Augmented Protein** (4 units, n = 298)	Period 2 – **Usual Protein** (4 units, n = 530)	Period 3 – **Augmented Protein** (4 units, n = 551)	Period 3 – **Usual Protein** (4 units, n = 368)	Period 4 – **Augmented Protein** (4 units, n = 352)	Period 4 – **Usual Protein** (4 units, n = 483)
Adverse Event, n (%)								
Number of participants with at least one event	0 (0.0)	0 (0.0)	1 (0.3)	0 (0.0)	2 (0.4)	1 (0.3)	0 (0.0)	0 (0.0)
Number of events	0	0	1	0	2	1	0	0
Serious Adverse Event, n (%)								
Number of participants with at least one event	1 (0.2)	0 (0.0)	0 (0.0)	1 (0.2)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
Number of events	1	0	0	1	0	0	0	0

eTable 14: Readmissions by treatment group

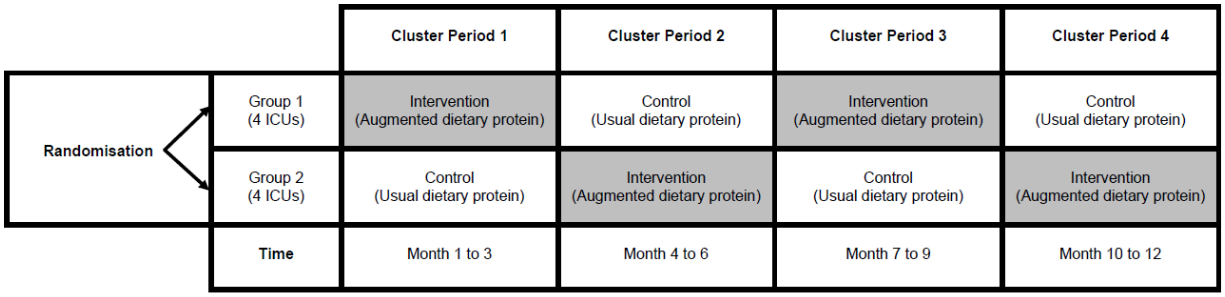
	Augmented Protein (n = 1681)	Usual Protein (n = 1716)
Readmissions (Total ICU and Hospital readmissions prior to Day 90), n (%)		
0	1,437 (85.5)	1,466 (85.4)
1	175 (10.4)	183 (10.7)
2	56 (3.3)	48 (2.8)
3	9 (0.5)	16 (0.9)
4	3 (0.2)	2 (0.1)
5	1 (0.1)	1 (0.1)
ICU readmissions (Readmissions to ICU within the index hospital admission, prior to Day 90), n (%)		
0	1,592 (94.7)	1,630 (95.0)
1	79 (4.7)	72 (4.2)
2	6 (0.4)	10 (0.6)
3	3 (0.2)	3 (0.2)
4	1 (0.1)	0 (0.0)
5	0 (0.0)	1 (0.1)
ICU readmissions (Readmissions to ICU post-index hospital admission, prior to Day 90[1]), n (%)		
0	1,646 (97.9)	1,693 (98.7)
1	33 (2.0)	20 (1.2)
2	1 (0.1)	2 (0.1)
3	1 (0.1)	1 (0.1)
Hospital readmissions (Readmissions to hospital post index hospital admission, prior to Day 90), n (%)		
0	1,520 (90.4)	1,544 (90.0)
1	139 (8.3)	142 (8.3)
2	18 (1.1)	25 (1.5)
3	3 (0.2)	5 (0.3)
5	1 (0.1)	0 (0.0)

Abbreviations: ICU = Intensive Care Unit

[1]Index hospital admission refers to the hospital admission in which the patient commenced trial enteral formula

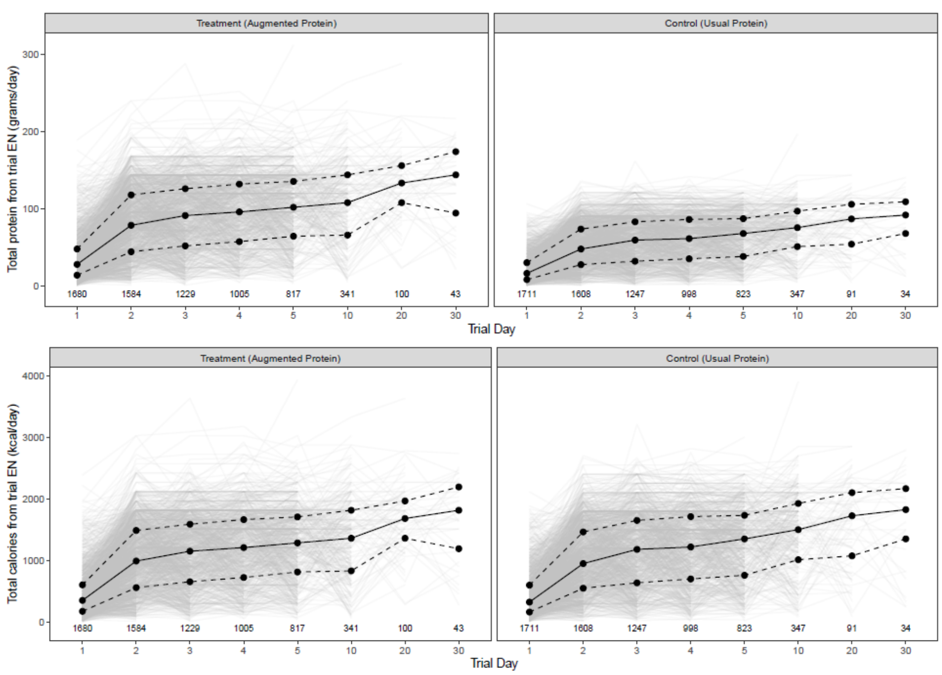
eFigures

eFigure 1: Study design



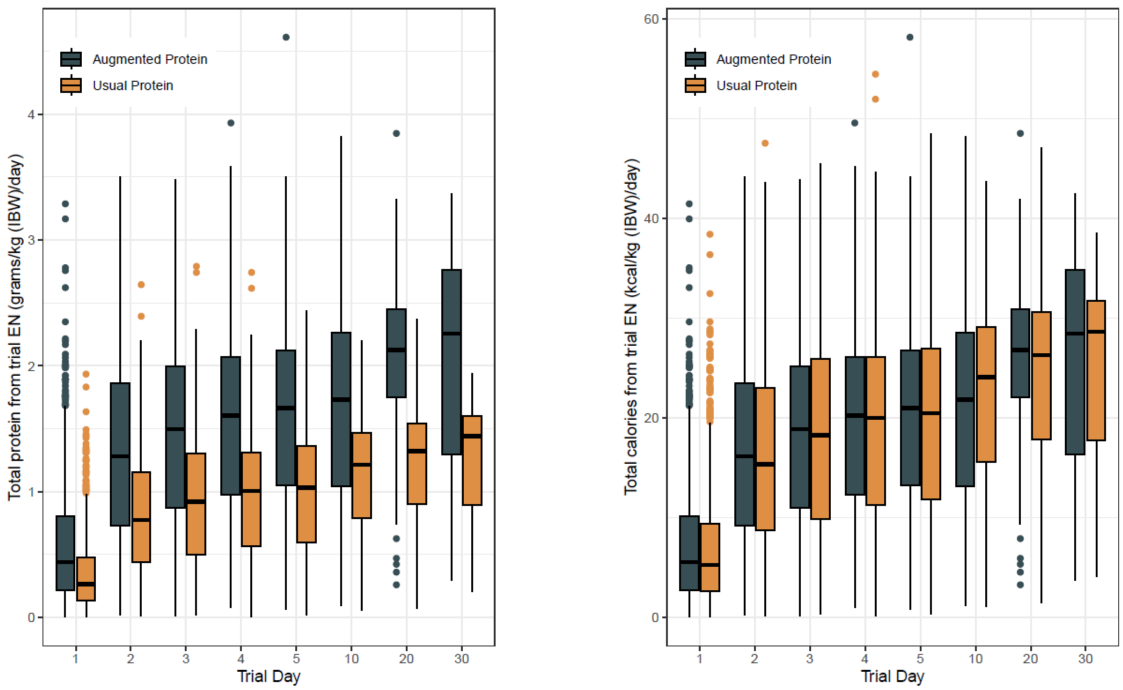
(Reprinted from *Critical Care and Resuscitation*, by Summers et al Vol 25, 2023, p.148. Copyright 2023 by Elsevier. Reprinted with permission from Editor-in-Chief for the purpose of scientific publication)[16].

eFigure 2: The between patient variability in protein and calories from trial enteral formula



Protein from trial nutrition – total g/day (top row). Calories from trial nutrition – total kcal/day (bottom row). Black solid line is the median and lower and upper dashed lines are the 25th and 75th percentiles, respectively. Grey solid lines are individual profiles.

eFigure 3: Protein delivered in grams per kilogram and calories delivered in kilocalories per kilogram of ideal body weight per day



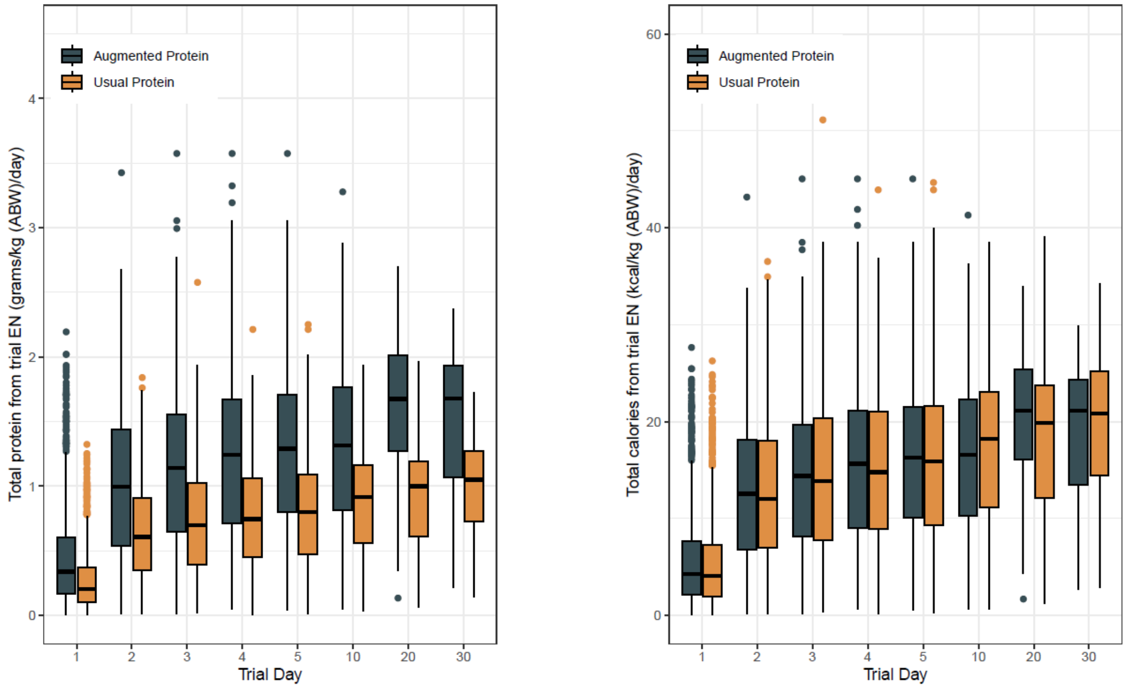
Number of Patients

Trial Day	1	2	3	4	5	10	20	30
Augmented Protein	1311	1243	966	799	658	283	84	34
Usual Protein	1301	1224	950	759	623	280	75	26

Protein from trial nutrition – total g/kg of ideal body weight/day (left) and calories from trial enteral nutrition – total kcal/ kg of ideal body weight day (right). Horizontal lines in boxes represent medians; bottoms of boxes show 25th percentile and tops of boxes show 75th percentile.

EN= enteral nutrition; IBW = Ideal body weight

eFigure 4: Protein delivered in grams per kilogram and calories delivered in kilocalories per kilogram of actual body weight per day



Number of Patients

Augmented Protein	1465	1382	1066	867	708	301	89	37
Usual Protein	1463	1375	1068	849	701	307	80	27

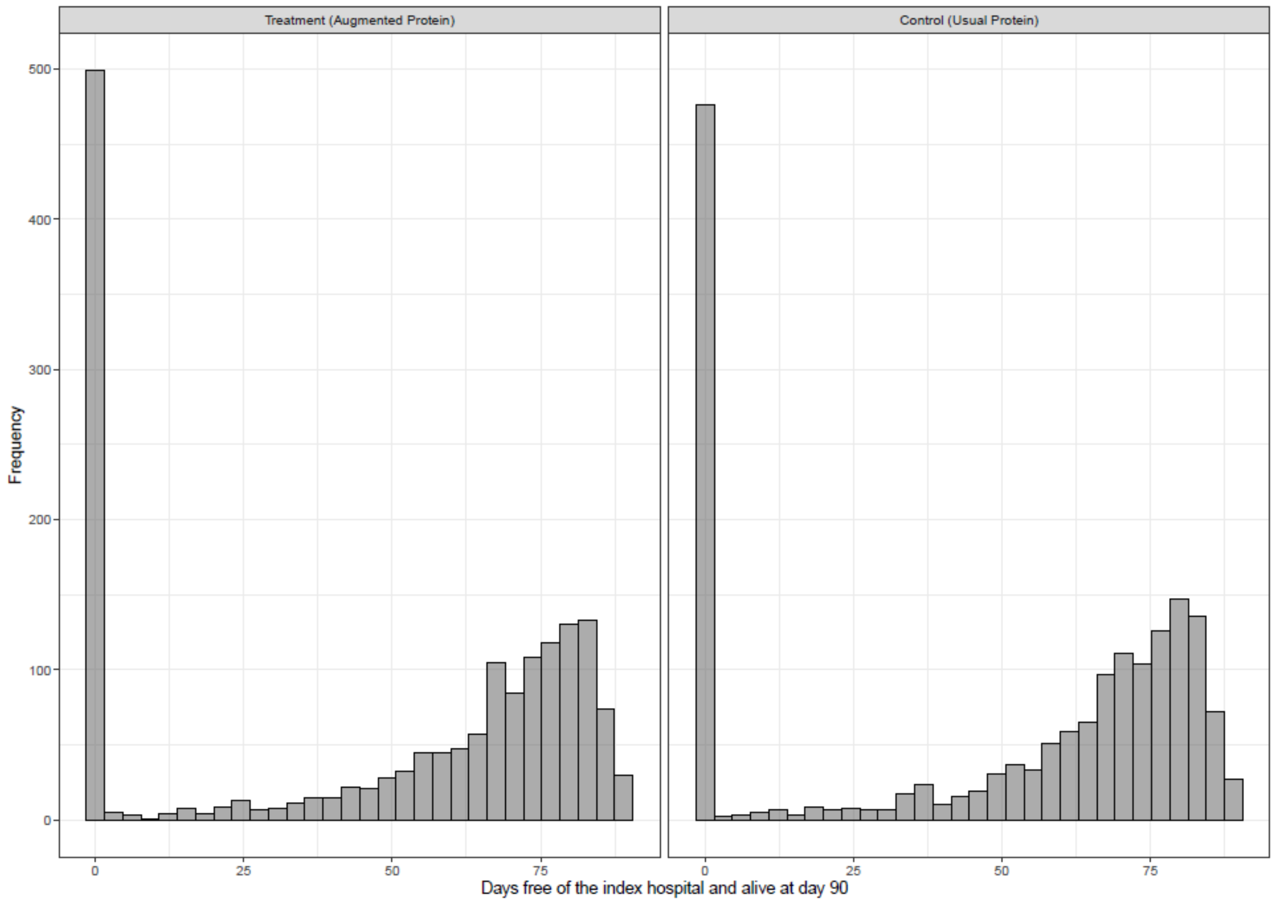
Number of Patients

Augmented Protein	1465	1382	1066	867	708	301	89	37
Usual Protein	1463	1375	1068	849	701	307	80	27

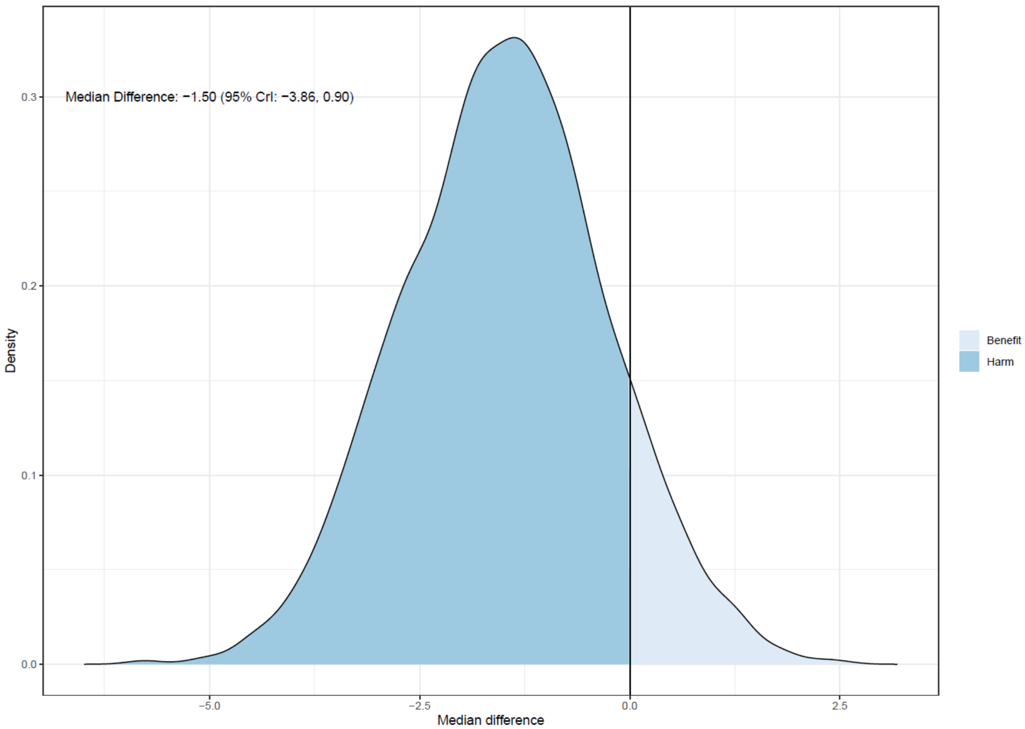
Protein from trial nutrition – total g/kg of actual body weight/day (left) and calories from trial enteral nutrition – total kcal/ kg of actual body weight day (right). Horizontal lines in boxes represent medians; bottoms of boxes show 25th percentile and tops of boxes show 75th percentile.

ABW = Actual body weight; EN= enteral nutrition

eFigure 5: Histogram of primary outcome by treatment group

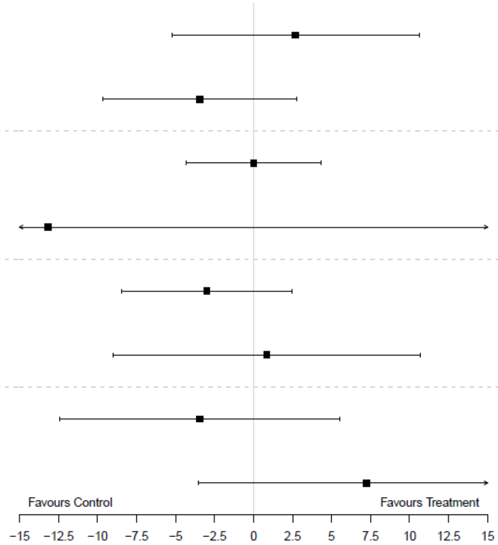


eFigure 6: Bayesian model analysis



eFigure 7: Forest Plot of Sub-Group Analysis

	N (Treatment)	Median (IQR) (Treatment)	N (Control)	Median (IQR) (Control)	Median difference (95% CI)	
IMV=No	296	69.0 (46.0−81.0)	361	67.0 (34.0−77.0)	2.7 (−5.24, 10.64)	p=0.023
IMV=Yes	1385	60.0 (0.0−76.0)	1355	63.0 (0.0−77.0)	−3.44 (−9.64, 2.76)	
RRT=No	1559	65.0 (0.0−78.0)	1597	65.0 (0.0−78.0)	0 (−4.32, 4.32)	p<0.001
RRT=Yes	122	15.0 (0.0−59.8)	119	24.0 (0.0−64.0)	−13.19 (−50, 23.62)	
Age<70	1231	65.0 (0.0−77.5)	1211	67.0 (24.0−79.0)	−3 (−8.48, 2.49)	p=0.106
Age>=70	450	53.5 (0.0−75.0)	505	51.0 (0.0−73.0)	0.85 (−8.98, 10.68)	
BMI<35	1066	62.0 (0.0−76.0)	1074	65.0 (0.0−77.0)	−3.46 (−12.44, 5.53)	p=0.468
BMI>=35	222	65.5 (8.5−77.0)	206	59.5 (0.0−75.8)	7.23 (−3.56, 18.02)	



Instrument for assessing the Credibility of Effect Modification Analyses (ICEMAN) in randomized controlled trials

Instrument for assessing the Credibility of Effect Modification Analyses (ICEMAN) in randomized controlled trials

Quick Instructions

- Synonyms for effect modification include subgroup effect, interaction, and moderation
- The instrument applies to a *single* proposed effect modification at a time; complete one form per each outcome, time-point, effect measure, and effect modifier
- Response options on the left indicate definitely or probably reduced, response options on the right probably or definitely increased credibility
- Completely unclear goes under probably reduced credibility
- It is helpful to provide a supporting comment or quotation under each question
- Whether an effect modification is patient-important is *not* part of the credibility assessment
- The manual provides more detailed instructions and examples

Preliminary considerations

Study reference(s): **TARGET Protein - heterogeneity of effect in patients receiving mechanical ventilation**

If available, protocol reference(s):

State a single outcome and, if applicable, time-point of interest (e.g. mortality at 1 year follow-up): **Days free of hospital and alive at day 90**

State a single effect measure of interest (e.g. relative or absolute risk difference): **Median (95 % confidence intervals)**

State a single potential effect modifier of interest (e.g. age or comorbidity): **Mechanical ventilation**

Was the potential effect modifier measured before randomization? [*] yes, continue [ ] no, stop here, refer to manual for further instructions

Credibility assessment

1: Was the direction of the effect modification correctly hypothesized a priori?

[*] Definitely no	[ ] Probably no or unclear	[ ] Probably yes	[ ] Definitely yes
Clearly post-hoc or results inconsistent with hypothesized direction or biologically very implausible	*Vague hypothesis or hypothesized direction unclear*	*No prior protocol available but unequivocal statement of a priori hypothesis with correct direction of effect modification*	*Prior protocol available and includes correct specification of direction of effect modification, e.g. based on a biologic rationale*

Comment: Hypothesized greater benefit to be observed in those receiving mechanical ventilation at baseline

2: Was the effect modification supported by prior evidence?

[ ] Inconsistent with prior evidence	**[*] Little or no support or unclear**	[ ] Some support	[ ] Strong support
Prior evidence suggested a different direction of effect modification	*No prior evidence or consistent with weak or very indirect prior evidence (e.g. animal study at high risk of bias) or unclear*	*Consistent with more limited or indirect prior evidence (e.g. large observational study, non-significant effect modification in prior RCT, or different population)*	*Consistent with strong prior evidence directly applicable to the clinical scenario (e.g. significant effect modification in related RCT)*

Comment: No existing data to support

3: Does a test for interaction suggest that chance is an unlikely explanation of the apparent effect modification? (consider irrespective of number of effect modifiers)

[ ] Chance a very likely explanation	**[*] Chance a likely explanation** or unclear	[ ] Chance may not explain	[ ] Chance an unlikely explanation
Interaction p-value >0.05	*Interaction p-value ≤0.05 and >0.01, or no test of interaction reported and not computable*	*Interaction p-value ≤0.01 and >0.005*	*Interaction p-value ≤0.005*

Comment: P=0.023

4: Did the authors test only a small number of effect modifiers or consider the number in their statistical analysis?

[ ] Definitely no	[ ] Probably no or unclear	**[*] Probably yes**	[ ] Definitely yes
Explicitly exploratory analysis or large number of effect modifiers tested (e.g. greater than 10) and multiplicity not considered in analysis	*No mention of number or 4-10 effect modifiers tested and number not considered in analysis*	*No protocol available but unequivocal statement of 3 or fewer effect modifiers tested*	*Protocol available and 3 or fewer effect modifiers tested or number considered in analysis*

Comment: **Four effect modifiers tested.**

Appendix to: Schandelmaier S, Briel M, Varadhan R, et al. Development of the Instrument to assess the Credibility of Effect Modification Analyses (ICEMAN) in randomized controlled trials and meta-analyses. *CMAJ* 2020. doi: 10.1503/cmaj.200077.

5: If the effect modifier is a continuous variable, were arbitrary cut points avoided? [*] not applicable: not continuous

[ ] Definitely no	[ ] Probably no or unclear	[ ] Probably yes	[ ] Definitely yes
Analysis based on exploratory cut point (e.g. picking cut point associated with highest interaction p-value)	*Analysis based on cut point(s) of unclear origin*	*Analysis based on pre-specified cut points, e.g. suggested by prior RCT*	*Analysis based on the full continuum, e.g. assuming a linear or logarithmic relationship*

Comment:

6 Optional: Are there any additional considerations that may increase or decrease credibility? (manual section 2.6)

[ ] yes, probably decrease [ ] yes, probably increase

Comment: **Not applicable**

7: How would you rate the overall credibility of the proposed effect modification?

The overall rating should be driven by the items that decrease credibility. The following provides a sensible strategy:

- All responses definitely or probably reduced credibility or unclear → very low
- Two or more responses definitely reduced credibility → maximum usually low even if all other responses satisfy credibility criteria
- One response definitely reduced credibility → maximum usually moderate even if all other responses satisfy credibility criteria
- Two responses probably reduced credibility → maximum usually moderate even if all other responses satisfy credibility criteria
- No response options definitely or probably reduced credibility → high very likely

Place a mark on the continuous line (e.g. hit "x" in electronic version)

X

Very low credibility	**Low credibility**	**Moderate credibility**	**High credibility**
Very likely no effect modification Use overall effect for each subgroup	Likely no effect modification Use overall effect for each subgroup but note remaining uncertainty	Likely effect modification Use separate effects for each subgroup but note remaining uncertainty	Very likely effect modification Use separate effects for each subgroup

Comment: **Deemed very low credibility of effect modification**

Appendix 5 (as supplied by the authors): Instrument for assessing the Credibility of Effect Modification Analyses (ICEMAN) in randomized controlled trials

Quick Instructions

- Synonyms for effect modification include subgroup effect, interaction, and moderation
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- Response options on the left indicate definitely or probably reduced, response options on the right probably or definitely increased credibility
- Completely unclear goes under probably reduced credibility
- It is helpful to provide a supporting comment or quotation under each question
- Whether an effect modification is patient-important is *not* part of the credibility assessment
- The manual provides more detailed instructions and examples

Preliminary considerations

Study reference(s): **TARGET Protein - heterogeneity of effect in patients with acute kidney failure**

If available, protocol reference(s):

State a single outcome and, if applicable, time-point of interest (e.g. mortality at 1 year follow-up): **Days free of hospital and alive at day 90**

State a single effect measure of interest (e.g. relative or absolute risk difference): **Median (95% confidence intervals)**

State a single potential effect modifier of interest (e.g. age or comorbidity): **Renal failure**

Was the potential effect modifier measured before randomization? **[*] yes, continue** [ ] no, stop here, refer to manual for further instructions

Credibility assessment

1: Was the direction of the effect modification correctly hypothesized a priori?

[ ] Definitely no	[ ] Probably no or unclear	**[*] Probably yes**	[ ] Definitely yes
Clearly post-hoc or results inconsistent with hypothesized direction or biologically very implausible	*Vague hypothesis or hypothesized direction unclear*	*No prior protocol available but unequivocal statement of a priori hypothesis with correct direction of effect modification*	*Prior protocol available and includes correct specification of direction of effect modification, e.g. based on a biologic rationale*

Comment: During conduct of TARGET Protein, the EFFORT Protein trial was published. Added renal failure as subgroup. Hypothesized th outcomes worse in those with kidney failure who received augmented protein

2: Was the effect modification supported by prior evidence?

[ ] Inconsistent with prior evidence	[ ] Little or no support or unclear	[ ] Some support	**[*] Strong support**
Prior evidence suggested a different direction of effect modification	*No prior evidence or consistent with weak or very indirect prior evidence (e.g. animal study at high risk of bias) or unclear*	*Consistent with more limited or indirect prior evidence (e.g. large observational study, non-significant effect modification in prior RCT, or different population)*	*Consistent with strong prior evidence directly applicable to the clinical scenario (e.g. significant effect modification in related RCT)*

Comment: Yes, see above (EFFORT Protein)

3: Does a test for interaction suggest that chance is an unlikely explanation of the apparent effect modification? (consider irrespective of number of effect modifiers)

[ ] Chance a very likely explanation	[ ] Chance a likely explanation or unclear	[ ] Chance may not explain	**[*] Chance unlikely explanation**
Interaction p-value >0.05	*Interaction p-value ≤0.05 and >0.01, or no test of interaction reported and not computable*	*Interaction p-value ≤0.01 and >0.005*	*Interaction p-value ≤0.005*

Comment: P<0.001

4: Did the authors test only a small number of effect modifiers or consider the number in their statistical analysis?

[ ] Definitely no	[ ] Probably no or unclear	**[*] Probably yes**	[ ] Definitely yes
Explicitly exploratory analysis or large number of effect modifiers tested (e.g. greater than 10) and multiplicity not considered in analysis	*No mention of number or 4-10 effect modifiers tested and number not considered in analysis*	*No protocol available but unequivocal statement of 3 or fewer effect modifiers tested*	*Protocol available and 3 or fewer effect modifiers tested or number considered in analysis*

Comment: Four effect modifiers tested

Appendix to: Schandelmaier S, Briel M, Varadhan R, et al. Development of the Instrument to assess the Credibility of Effect Modification Analyses (ICEMAN) in randomized controlled trials and meta-analyses. *CMAJ* 2020. doi: 10.1503/cmaj.200077.

5: If the effect modifier is a continuous variable, were arbitrary cut points avoided? [*] **not applicable: not continuous**

[ ] Definitely no	[ ] Probably no or unclear	[ ] Probably yes	[ ] Definitely yes
Analysis based on exploratory cut point (e.g. picking cut point associated with highest interaction p-value)	*Analysis based on cut point(s) of unclear origin*	*Analysis based on pre-specified cut points, e.g. suggested by prior RCT*	*Analysis based on the full continuum, e.g. assuming a linear or logarithmic relationship*

Comment:

6 Optional: Are there any additional considerations that may increase or decrease credibility? (manual section 2.6)

[*] yes, probably decrease [ ] yes, probably increase

Comment: **Difference in definition with commencement of new renal replacement therapy prior to commencing study used for TARGET Protein. Also small numbers in the cohort (augmented protein=122, usual protein=119 and wide 95% CI (-50.00, +23.62) days**

7: How would you rate the overall credibility of the proposed effect modification?
The overall rating should be driven by the items that decrease credibility. The following provides a sensible strategy:

- All responses definitely or probably reduced credibility or unclear → very low
- Two or more responses definitely reduced credibility → maximum usually low even if all other responses satisfy credibility criteria
- One response definitely reduced credibility → maximum usually moderate even if all other responses satisfy credibility criteria
- Two responses probably reduced credibility → maximum usually moderate even if all other responses satisfy credibility criteria
- No response options definitely or probably reduced credibility → high very likely

Place a mark on the continuous line (e.g. hit "x" in electronic version)

X

Very low credibility	**Low credibility**	**Moderate credibility**	**High credibility**
Very likely no effect modification Use overall effect for each subgroup	Likely no effect modification Use overall effect for each subgroup but note remaining uncertainty	Likely effect modification Use separate effects for each subgroup but note remaining uncertainty	Very likely effect modification Use separate effects for each subgroup

Comment:

Deemed moderate credibility of effect modification

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